
DSC5001: Statistical Machine Learning I

Tutorial 3: Bootstrap and Statistical Uncertainty

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Tutorial 3

Complete Tasks 1–3 and show the main steps of your calculation.

- ▶ Task 1: Sampling uncertainty and sample-size planning
- ▶ Task 2: Exact bootstrap distribution by hand
- ▶ Task 3: Paired bootstrap for model comparison
- ▶ Optional Task 4: Bootstrap simulation in Python

For each task, connect the numerical result to the sampling question it answers.

Task 1: Bernoulli Uncertainty and Sample Size

Let:

$$X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p), \quad \hat{p} = \frac{1}{n} \sum_{i=1}^n X_i.$$

1. Derive $E[\hat{p}]$ and $Var(\hat{p})$.
2. Hence derive the standard error of $\hat{p}$.
3. Suppose $n = 100$ and 30 successes are observed. Estimate the standard error.
4. Using $\hat{p} = 0.30$ as a planning value, find the minimum n required for estimated standard error at most 0.02.
5. If no preliminary value of p is available, use the worst case $p(1 - p) \leq 1/4$ to find a conservative sample size.

Solution: Task 1 — Sampling Distribution

Because $E[X_i] = p$:

$$E[\hat{p}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n p = \boxed{p}.$$

Using independence and $Var(X_i) = p(1 - p)$:

$$Var(\hat{p}) = Var\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n p(1 - p) = \boxed{\frac{p(1 - p)}{n}}.$$

Therefore:

$$\boxed{se(\hat{p}) = \sqrt{\frac{p(1 - p)}{n}}}$$

So $\hat{p}$ is unbiased, while its uncertainty decreases at the $1/\sqrt{n}$ rate.

Solution: Task 1 — Planning the Sample Size

With $n = 100$ and $\hat{p} = 0.30$:

$$\widehat{\text{se}}(\hat{p}) = \sqrt{\frac{0.30(0.70)}{100}} \approx \boxed{0.0458}.$$

For target standard error $r = 0.02$:

$$\sqrt{\frac{0.30(0.70)}{n}} \leq 0.02$$

implies:

$$n \geq \frac{0.30(0.70)}{0.02^2} = \boxed{525}.$$

Without a planning value for p :

$$p(1 - p) \leq \frac{1}{4}.$$

Hence:

$$n \geq \frac{1/4}{0.02^2} = \boxed{625}.$$

Task 2: Exact Bootstrap Distribution

Consider the observed sample:

$$D = [2, 4].$$

The statistic of interest is the sample mean:

$$\hat{\theta} = \bar{x} = 3.$$

A bootstrap sample contains two draws with replacement from D .

1. List all possible ordered bootstrap samples.
2. Compute the bootstrap mean $\bar{X}^*$ for each sample.
3. Derive the exact probability distribution of $\bar{X}^*$.
4. Compute $E_*(\bar{X}^*)$, the bootstrap bias, and $Var_*(\bar{X}^*)$.
5. Compare the bootstrap SE with the analytic estimate $s/\sqrt{n}$.

Solution: Task 2 — Exact Distribution

There are $2^2 = 4$ equally likely ordered bootstrap samples:

Bootstrap sample	$\bar{X}^*$	Probability
[2, 2]	2	1/4
[2, 4]	3	1/4
[4, 2]	3	1/4
[4, 4]	4	1/4

Therefore:

$$P_*(\bar{X}^* = 2) = \frac{1}{4}, \quad P_*(\bar{X}^* = 3) = \frac{1}{2}, \quad P_*(\bar{X}^* = 4) = \frac{1}{4}.$$

The bootstrap mean is:

$$E_*(\bar{X}^*) = 2 \left(\frac{1}{4} \right) + 3 \left(\frac{1}{2} \right) + 4 \left(\frac{1}{4} \right) = \boxed{3}.$$

Thus:

$$\widehat{\text{Bias}}_{\text{boot}} = E_*(\bar{X}^*) - \bar{x} = \boxed{0}.$$

Solution: Task 2 — Bootstrap Standard Error

Using $E_*(\bar{X}^*) = 3$:

$$Var_*(\bar{X}^*) = \frac{1}{4}(2-3)^2 + \frac{1}{2}(3-3)^2 + \frac{1}{4}(4-3)^2 = \boxed{\frac{1}{2}}.$$

Hence:

$$se_{boot}(\bar{X}) = \sqrt{1/2} \approx \boxed{0.707}.$$

For the original data:

$$s^2 = \frac{(2-3)^2 + (4-3)^2}{2-1} = 2,$$

so the usual analytic estimate is:

$$\widehat{se}_{analytic} = s/\sqrt{2} = \boxed{1}.$$

For $n = 2$, the exact bootstrap relation is:

$$se_{boot} = \widehat{se}_{analytic} \sqrt{\frac{n-1}{n}} = 1\sqrt{\frac{1}{2}} = 0.707.$$

Task 3: Paired Bootstrap for Two Classifiers

Two classifiers are evaluated on the same eight test examples.

Example i	1	2	3	4	5	6	7	8
Model A correct?	1	1	1	1	0	1	0	1
Model B correct?	1	1	0	1	0	0	1	1

Let:

$$\hat{\Delta} = \widehat{\text{Acc}}_A - \widehat{\text{Acc}}_B.$$

1. Compute both test accuracies and $\hat{\Delta}$.
2. One bootstrap resample uses indices: $[1, 2, 2, 3, 5, 6, 8, 8]$. Compute $\hat{\Delta}^*$.
3. Five bootstrap differences are:

$$0, \quad 0.125, \quad 0.125, \quad 0.250, \quad 0.125.$$

Compute their mean, estimated bootstrap bias, and bootstrap SE.

4. Why must the same resampled indices be used for both models?

Solution: Task 3 — Original and Bootstrap Difference

Model A is correct on 6 of 8 examples:

$$\widehat{\text{Acc}}_A = \frac{6}{8} = 0.750.$$

Model B is correct on 5 of 8 examples:

$$\widehat{\text{Acc}}_B = \frac{5}{8} = 0.625.$$

Therefore:

$$\widehat{\Delta} = 0.750 - 0.625 = 0.125$$

For bootstrap indices [1, 2, 2, 3, 5, 6, 8, 8]:

$$\widehat{\text{Acc}}_A^* = \frac{7}{8} = 0.875, \quad \widehat{\text{Acc}}_B^* = \frac{5}{8} = 0.625.$$

Thus:

$$\widehat{\Delta}^* = 0.875 - 0.625 = 0.250$$

Solution: Task 3 — Bias and Standard Error

The five bootstrap differences are:

$$0, \quad 0.125, \quad 0.125, \quad 0.250, \quad 0.125.$$

Their mean is:

$$\widehat{\Delta}^* = \frac{0 + 0.125 + 0.125 + 0.250 + 0.125}{5} = 0.125.$$

Therefore:

$$\widehat{\text{Bias}}_{\text{boot}} = 0.125 - 0.125 = 0$$

The bootstrap SE is:

$$\widehat{\text{se}}_{\text{boot}} = \sqrt{\frac{(0 - 0.125)^2 + (0.250 - 0.125)^2}{4}} \approx \boxed{0.088}.$$

Both models must use the same resampled indices because their predictions are paired on the same test examples.

Optional Task 4: Bootstrap Simulation in Python

Generate one fixed right-skewed sample:

$$X_1, \dots, X_{80} \stackrel{i.i.d.}{\sim} \text{Exponential}(1).$$

Use random seed 5001 for the observed sample and seed 5002 for bootstrap resampling.
With $B = 5000$ bootstrap samples:

1. Compute the observed sample mean and median.
2. Estimate the bootstrap SE of the mean and median.
3. Construct 95% percentile intervals for both statistics.
4. Compare the bootstrap SE of the mean with $s/\sqrt{n}$.
5. Plot the bootstrap distributions of the mean and median.

Explain why the bootstrap is especially useful for the median.

Solution: Optional Task 4 — Python I

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(5001)
x = rng.exponential(scale=1.0, size=80)

mean_hat = np.mean(x)
median_hat = np.median(x)
analytic_se = np.std(x, ddof=1) / np.sqrt(len(x))

rng_boot = np.random.default_rng(5002)
B = 5000
idx = rng_boot.integers(0, len(x), size=(B, len(x)))
boot = x[idx]

boot_means = np.mean(boot, axis=1)
boot_medians = np.median(boot, axis=1)
```

Solution: Optional Task 4 — Python II

```
mean_se = np.std(boot_means, ddof=1)
median_se = np.std(boot_medians, ddof=1)

mean_ci = np.quantile(boot_means, [0.025, 0.975])
median_ci = np.quantile(boot_medians, [0.025, 0.975])

print("Mean:", mean_hat)
print("Median:", median_hat)
print("Analytic SE:", analytic_se)
print("Bootstrap mean SE:", mean_se)
print("Bootstrap median SE:", median_se)
print("Mean 95% CI:", mean_ci)
print("Median 95% CI:", median_ci)
```

Solution: Optional Task 4 — Results

The numerical results are approximately:

$$\hat{\mu} = 0.9721, \quad \widehat{\text{median}} = 0.5938.$$

For the mean:

$$\hat{\text{se}}_{\text{analytic}} = 0.1107, \quad \hat{\text{se}}_{\text{boot}} = 0.1124,$$

and:

$$CI_{0.95}^{\text{mean}} \approx [0.7618, 1.2040]$$

For the median:

$$\hat{\text{se}}_{\text{boot}} = 0.1416,$$

with:

$$CI_{0.95}^{\text{median}} \approx [0.4413, 0.9335]$$

The mean bootstrap SE is close to the analytic result, while the median uncertainty is obtained without deriving a separate analytic sampling formula.

Solution: Optional Task 4 — Visualization

```
plt.figure()
plt.hist(boot_means, bins=35)
plt.axvline(mean_hat, linewidth=2)
plt.xlabel("Bootstrap mean")
plt.ylabel("Count")
plt.title("Bootstrap Distribution of the Mean")
plt.show()
```

```
plt.figure()
plt.hist(boot_medians, bins=35)
plt.axvline(median_hat, linewidth=2)
plt.xlabel("Bootstrap median")
plt.ylabel("Count")
plt.title("Bootstrap Distribution of the Median")
plt.show()
```

Tutorial 3: derive, calculate, and verify bootstrap uncertainty

Questions and discussion